

Standard Form & Indices

Test 6 · Answer sheet · 41 marks total

Section A — Laws of indices

A1. 7^9 (1)

A2. 5^5 (1)

A3. 3^8 (1)

A4. $2^4 \times 2^2 = 2^6$ (1) = **64** (1)

Section B — Zero and negative indices

B1. 1 (1)

B2. $1 \div 7^2 = 1 \div 49$ (1) = **1/49** (1)

B3. $1 \div 10^7 = 1 \div 10000000$ (1) = **0.0000001** (1)

Section C — Writing standard form

C1. 3.6 (1) = **3.6×10^4** (1)

C2. 7.4 (1) = **7.4×10^6** (1)

C3. 5.2 (1) = **5.2×10^{-3}** (1)

C4. 7 (1) = **7×10^{-4}** (1)

Section D — Converting back, and calculating

D1. **8,300** (1)

D2. **0.000002** (1)

D3. $7 \times 3 = 21$, $10^3 \times 10^5 = 10^8$ (1) = 21×10^8 (1) adjust to standard form: **2.1×10^9** (1)

D4. $3 \div 6 = 0.5$, $10^9 \div 10^4 = 10^5$ (1) = 0.5×10^5 (1) adjust to standard form: **5×10^4** (1)

Section E — Powers to know, and estimating roots

E1. **256** (1)

E2. $7^2=49$, $8^2=64$ = between **7 and 8** (1)

Section F — Laws of indices with algebra

F1. x^{12} (1)

F2. y^5 (1)

F3. z^{12} (1)

F4. $3 \times 8 = 24$ (1) $m^{5+2} = m^7$ (1) = **$24m^7$**

F5. $4^3 = 64$ (1) $n^{2 \times 3} = n^6$ (1) = **$64n^6$**

Section G — Fractional indices (Higher)

G1. $\sqrt{100} = \mathbf{10}$ (1)

G2. cube root of 216 = **6** (1)

G3. fifth root of 32 = 2 (1) $2^3 = \mathbf{8}$ (1)

G4. $\sqrt{36} = 6$ (1) $1 \div 6 = \mathbf{1/6}$ (1)

Marking note: award method marks for a correctly identified front number/power, or correctly applied law of indices, even if the final answer contains an arithmetic error. Accept equivalent fractions/decimals.